Exp 7a - Calorimetry

INTRODUCTION

Calorimetry is a technique used to measure heat given off or absorbed by a process. Since it is not possible to measure heat directly, calorimeters generally measure temperature changes, which are related to heat by the following equation:

$$q = m \cdot c \cdot \Delta T \quad (6)$$

q = heat, usually measured in J or cal

m = mass of the substance

c = specific heat capacity of the substance

ΔT = change in temperature in °C (measured as the final temperature – initial temperature)

The physical change or chemical change (reaction) is generally considered the "system", while the solution in which the reaction takes place would be considered the "surroundings". In the calorimeter setup for this experiment, the thermometer is placed in the solution (surroundings). Since under these conditions, significant heat is not gained or lost, but only transferred from one thing to another, the sum must equal zero, as shown in the equation below.

$$q_{system} + q_{surroundings} = 0 \quad (7)$$

SAFETY EQUIPMENT REQUIRED FOR LAB

- Goggles
- Lab coat
- Nitrile gloves (provided)

SUPPLY LIST

- Calorimeter (1 Styrofoam cup, 1 plastic cup, 1 cardboard lid, thermometer)
- Bench Post
- 250-mL or larger beaker (temporary waste beaker)
- Thermometer clamp
- 100-mL graduated cylinder
- Scoopula
- Wash Bottle filled with DI Water

CLASSROOM SUPPLY AND REACTANTS

- Citric acid (solid)
- Sodium Carbonate (solid)
- DI Water
- Ice

PART 1 – Exothermic and Endothermic Changes

BACKGROUND

Many solids will absorb or release heat when they dissolve in liquids due to interactions with the solvent molecules. We will dissolve the solid citric acid ($H_3C_6H_5O_7$) and sodium carbonate (Na_2CO_3), separately, in water to observe this process.

Endothermic versus Exothermic

When a process releases heat, we call the process *exothermic*. (If an exothermic process happens in a beaker, it will feel warmer because it is giving off heat to your hand.)

When a process absorbs heat, we call the process *endothermic*. (If an endothermic process happens in a beaker, it will feel colder because it is absorbing heat from your hand.)

PROCEDURE

1. ☐ Set up a calorimeter as shown in figure 1. A plastic cup is used inside a Styrofoam cup to provide an extra insulation layer, which slows down the transfer of heat between the cup and the surroundings.
2. ☐ Obtain the mass of your empty calorimeter (a plastic cup inside a Styrofoam cup).

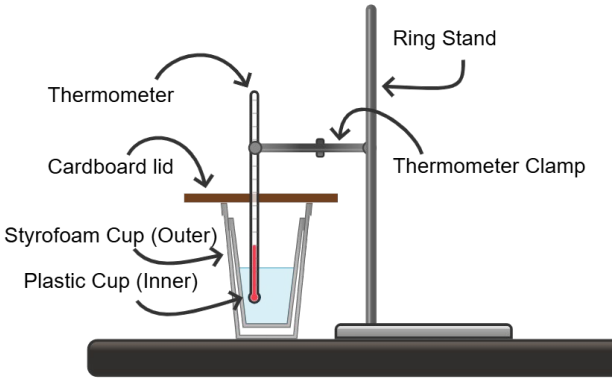


Figure 1. Calorimeter Setup

**** Do ***not*** include the mass of the cardboard lid in your measurement.

3. ☐ Using a graduated cylinder measure between 45.0-55.0 mL DI water. Pour into your calorimeter.
4. ☐ Record the initial temperature of the water to the nearest 0.1 °C. Write this temperature in the box for "Initial Temperature" in the column for $H_3C_6H_5O_7$ on the data report.
5. ☐ Weigh between 4.8-5.2 g of citric acid ($H_3C_6H_5O_7$) on a piece of weigh paper.
    a. Use a brush to sweep any spills inside the balance out. Wasted material can go into the aqueous inorganic waste container.
    b. Please ensure the bottle lid is screwed back on before leaving the balance area.
6. ☐ Add the citric acid to the calorimeter, then place the cardboard lid back onto the calorimeter, and *carefully* stir the mixture with the thermometer until all the solid dissolves.

7. ☐ Record the highest temperature (or lowest temperature if the temperature goes down) reached as the final temperature, to the nearest 0.1 ℃.
8. ☐ Record the mass of the calorimeter and the solution.
   **** Do ***not*** include the mass of the cardboard lid in your measurement.
9. ☐ Empty the contents of the calorimeter into a large temporary waste beaker, then thoroughly rinse and dry the inner plastic cup of your calorimeter.
10. ☐ Repeat steps 3-8 using sodium carbonate (Na_2CO_3) instead of citric acid.
11. ☐ Slowly empty the contents of the calorimeter into a large temporary waste beaker. Once waste from citric acid and sodium carbonate are combined, waste can be poured down the drain. **** Combining citric acid and sodium carbonate will produce bubbles (carbon dioxide gas).

PART 2 – Specific Heat Capacity and Enthalpy of Fusion

BACKGROUND

When two objects at different temperatures come into contact with each other, heat from the hotter object will transfer to the cooler object until they reach the same temperature. The first law of thermodynamics (law of conservation of energy) states, that "Energy is never created or destroyed." If we focus only on heat energy and ignore other types of energy, this could be stated as follows:

$$q_1 + q_2 + q_3 + q_4 + \ldots = 0 \quad (8)$$

In other words, the heat of one object might go up and the heat of one object might go down, but the total amount of heat will stay the same. In this experiment, we will be putting ice into water and seeing if the sum of the heats is zero.

Heat from the water will transfer to the ice according to the equation below:

$$q = m \cdot c \cdot \Delta T \quad (9)$$

The ice will absorb the heat lost by the water as it melts. The heat gained by the ice will be calculated as:

$$q = n\,\Delta H_{fus} \quad (12)$$

For water, $\Delta H_{fus} = 6.01\ kJ/mol$

PROCEDURE

1. ☐ Using a graduated cylinder measure 45.0-55.0 mL DI water.
2. ☐ Record the mass of your calorimeter containing water.
   **** Do ***not*** include the mass of the cardboard lid in your measurement.
3. ☐ Record the initial temperature of the water to the nearest 0.1 °C.
4. ☐ Bring your calorimeter, and cardboard lid over to the bin filled with ice.
5. ☐ Carefully transfer 2-3 squares of ice to your calorimeter. The squares of ice may be separate or frozen together, either is fine.
6. ☐ Place the lid back on the calorimeter and stir using the thermometer until all the ice melts. Record the lowest temperature reached to the nearest 0.1 °C as the final temperature.
7. ☐ Record the mass of the calorimeter containing the water and melted ice.
8. ☐ Empty the contents of the calorimeter (just water) down the sink drain.

CLEAN-UP AND WASTE DISPOSAL

1. ☐ Once you finish collecting all your data, return your calorimeter's cardboard lid and Styrofoam cup to the supply bin. Throw away the plastic cup in the trash can.

DATA AND REPORT – DUE ____ / ____ / ____ **SECTION _____**

* Show your work and keep track of significant figures. **DATE _____**

PARTNER _____

PART 1 – Exothermic and Endothermic Changes

DATA

Mass of calorimeter (empty): _____

	$H_3C_6H_5O_7$	Na_2CO_3
Initial Temperature (°C)		
Final Temperature (°C)		
Mass of calorimeter *and* solution (g)		

PART 2 – Specific Heat Capacity and Enthalpy of Fusion

DATA

	Data
Mass of calorimeter + water (g)	
Initial Temperature of water (°C)	
Final Temperature of water (°C)	
Mass of calorimeter + water + ice (g)	

CALCULATIONS – PART 1 – EXOTHERMIC AND ENDOTHERMIC CHANGES

* Show your work, complete with units for $H_3C_6H_5O_7$. I will assume you did Na_2CO_3 the same way

1. ☐ Mass of the solution

2. ☐ Change in Temperature of Solution

3. ☐ Heat of Solution ($q_{solution}$)

RESULTS – PART 1 – ENDOTHERMIC AND EXOTHERMIC CHANGES

	$H_3C_6H_5O_7$	Na_2CO_3
Mass of Solution (m)		
Specific Heat of Solution (c)	4.184 J/g°C	4.184 J/g°C
Change in Temperature of Solution(ΔT)		
Heat of solution ($q_{solution}$)		
Heat of reaction* ($q_{reaction} + q_{solution} = 0$)		
Reaction is: Exothermic or Endothermic		

* The water in the calorimeter is considered the reaction's surroundings, therefore if heat is released by the reaction, it is absorbed by the water and vice-versa.

CALCULATIONS – PART 2 – SPECIFIC HEAT CAPACITY AND ENTHALPY OF FUSION

* Show your work, complete with units.

1. Mass of H_2O

2. Change in Temperature of Water

3. q_{water}

4. Mass of Ice

5. Moles of Ice

6. q_{ice}

7. $q_{water} + q_{ice}$

RESULTS

	Results
Mass of H_2O	
Specific Heat of water (c)	4.184 J/g°C
Change in Temperature of water (ΔT) ($\Delta T = T_f - T_i$)	
q_{water} (kJ)	
Mass of Ice	
Moles of Ice (n)	
Enthalpy of Fusion (ΔH_{fus})	6.01 kJ/mol
q_{ice} (kJ)	
$q_{water} + q_{ice}$	

QUESTIONS

1. Is ice melting endothermic or exothermic? *Provide data or results from this experiment to support your claim.*

2. Was energy conserved (i.e. is $q_{water} + q_{ice} = 0$)? If not, provide at least one error specific to your experiment that could account for the difference.